

AI Agents in Production: Stop Exposing Your AI Directly

How Azure APIM makes AI agents cost-efficient, fair, and production-ready

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You shipped an AI agent. Then the bill arrived.



One tenant used 80% of your daily token quota

No rate limiting



10 users asked the same question, 10 OpenAI round trips

No caching



₹40,000 cloud bill. Zero idea who caused it.

No visibility

GPT-4o costs approx \$0.0025 per 1K output tokens. At 1M tokens/day across tenants, costs spiral fast without guardrails.

Exposing your AI agent endpoint directly is a liability, not a shortcut.

APIM is the production layer your AI agent doesn't have yet



Gateway

Centralised entry point – auth, routing, transformation



Policies

Code-free XML rules – rate limit, cache, rewrite, log



Developer Portal

Self-service API keys and docs for your tenants



Analytics

Built-in usage, latency, and error dashboards

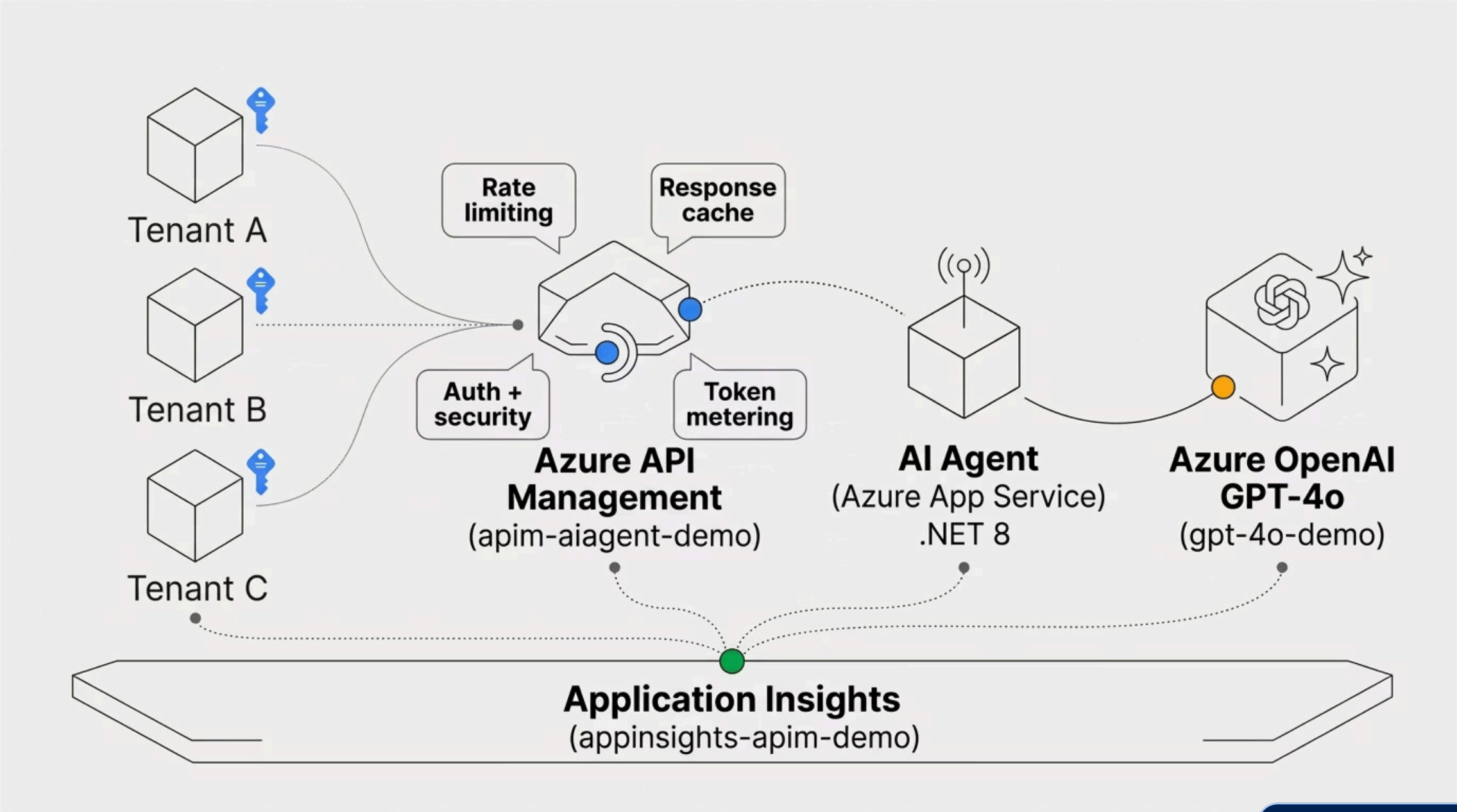
❏ Used by enterprises to manage millions of API calls daily. Now being used for AI agent workloads.

Azure API Management – generally available since 2014, 10+ years in production at scale

Made with **GAMMA**

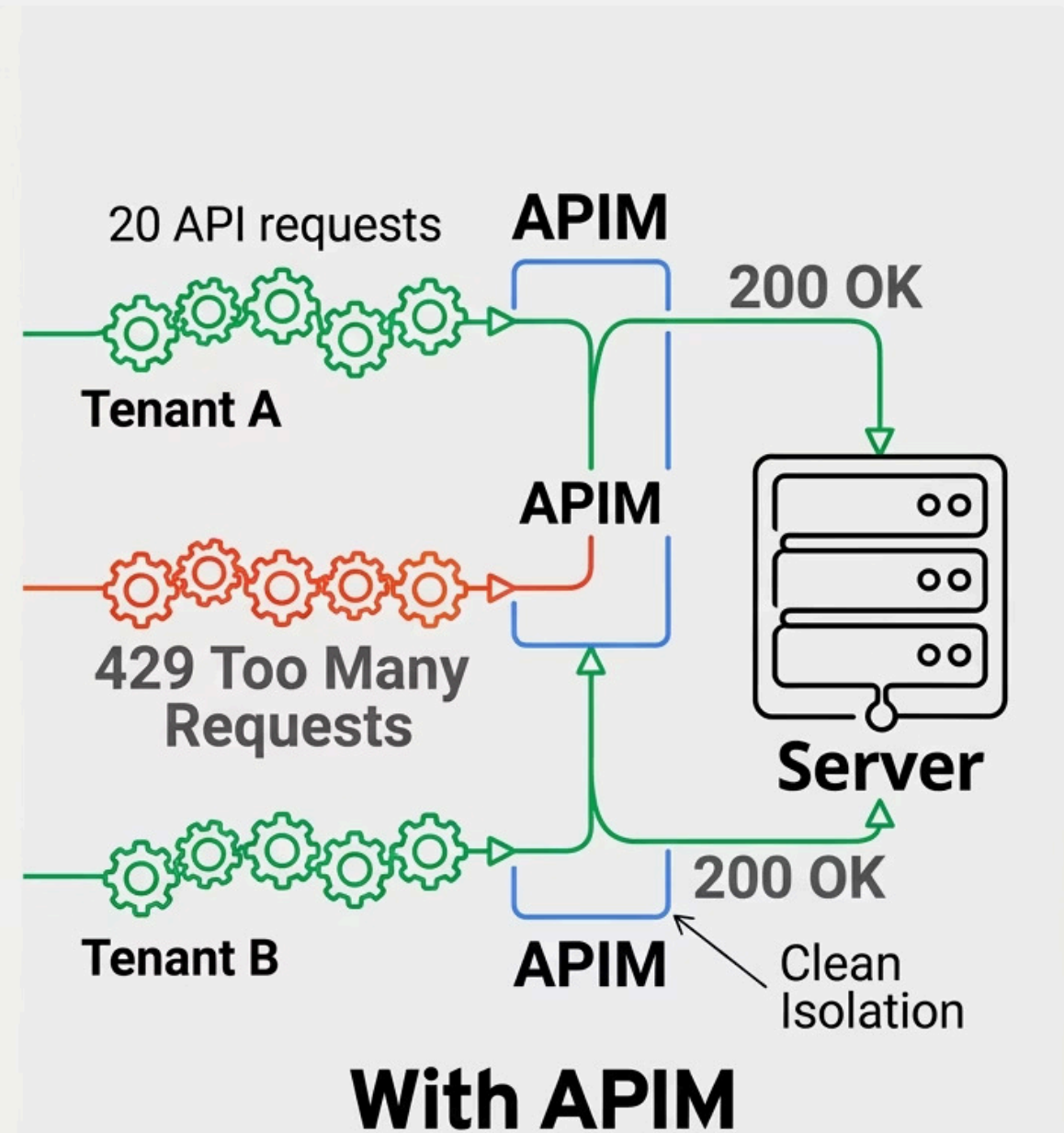
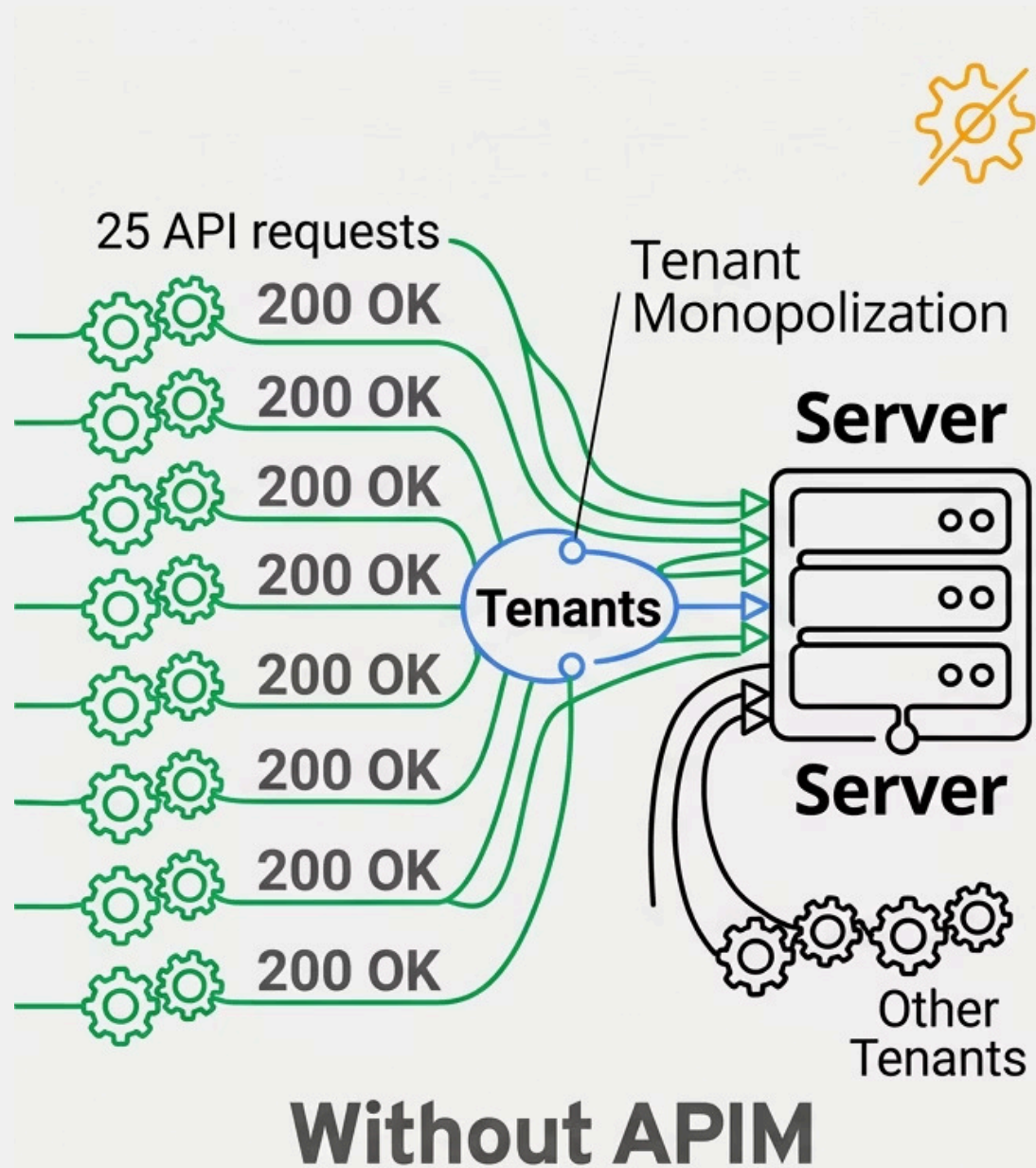
The Architecture We're Building Today

Every tenant goes through APIM. Nothing hits OpenAI directly.



Demo 1: Per-Tenant Rate Limiting

Tenant A hits the wall. Tenant B never notices.



```
<rate-limit-by-key
  calls="20"
  renewal-period="60"
  counter-key="@context.Subscription.Id"
  remaining-calls-header-name="X-RateLimit-Remaining" />
```

Per-tenant isolation. Defined by subscription key.

No code change in your .NET API. Policy only.

Changed in 5 minutes. Zero redeploy.

Made with GAMMA

Demo 2: Response Caching

The second identical prompt costs exactly zero tokens.

CACHE MISS (first call)

Response time: ~2,100ms

Tokens consumed: 297

OpenAI charged: Yes

CACHE HIT (same prompt again)

Response time: ~18ms

Tokens consumed: 0

OpenAI charged: No

Header: X-Cache: HIT

1,000 repeated FAQ prompts/day × 297 tokens × \$0.0025/1K tokens = ~\$0.74/day saved from caching alone

At scale across 50 tenants: meaningful monthly saving with one XML policy block

```
<cache-store-value duration="300" />  
<cache-lookup-value />
```

Prompt-aware caching. No changes to your agent code.

Demo 3: Bonus — Model + Prompt Optimisation

Same question. Same quality. 50× cheaper.

Configuration	Completion tokens	Relative cost	Output quality
GPT-4o + weak system prompt	297 tokens	1× (baseline)	Markdown headers and 7 bullet points nobody asked for
GPT-4o-mini + weak system prompt	~260 tokens	~0.07× cost	Shorter, still unstructured
GPT-4o-mini + optimised system prompt	~70 tokens	~0.02× cost	Clean 2-sentence answer ✓

"You are a concise technical assistant. Answer in 2-3 plain sentences maximum.
No bullet points, no headers, no examples unless asked."

- ❏ **This system prompt is enforced at the APIM gateway. Tenants cannot override it. One policy change, every tenant, instantly.**

Token Visibility + Cost Accountability

Know exactly who spent what. In real time.

A single Application Insights query gives you full cost accountability per tenant – no custom billing code required.

Timestamp	Tenant Name	Total Tokens	Prompt Tokens	Completion Tokens	Cache Hit
14:22:01	Tenant-A	297	28	269	MISS
14:22:14	Tenant-A	0	0	0	HIT
14:22:31	Tenant-B	84	28	56	MISS

```
traces
| extend data = parse_json(message)
| project timestamp, tenantName = tostring(data.tenantName),
    totalTokens = toint(data.totalTokens), cacheHit = tostring(data.cacheHit)
```

📄 No custom billing code. No middleware. Telemetry emitted directly from APIM policy via `<trace>` to Application Insights.

What You Get for Free

Five production problems. One APIM policy file.



Tenant isolation

rate-limit-by-key on subscription ID



Cost control

response caching with configurable TTL



Security

subscription key auth, no direct endpoint exposure



Observability

token usage, cache hit rate, per-tenant analytics



Prompt governance

system prompt enforced at gateway, not in code



This replaces what most teams would build as 3-4 separate middleware services, each needing their own deployment pipeline.

Resources Used in This Demo



Resource Group: rg-apim-demo



Azure OpenAI: openai-apim-demo

- GPT-4o deployment
- GPT-4o-mini deployment



App Service: api-aiagent-demo

- .NET 8 Minimal API



API Management: apim-aiagent-demo

- Developer tier



Application Insights: appinsights-apim-demo

Tech Stack: .NET 8 · Azure App Service · Azure OpenAI SDK v2 · Azure Portal



APIM Developer tier costs approx \$50/month. Upgrade to Standard for SLA and production scale.

Three Things to Do This Week

You don't need to boil the ocean. Start here.

1

**Put APIM in front of your
AI agent endpoint**

even if you only enable auth for now

2

**Add rate-limit-by-key per
subscription**

this is the single highest-ROI policy

3

**Add a tight system prompt
and switch non-critical
paths to GPT-4o-mini**

biggest cost lever with no quality loss

Every policy in this demo is live XML you can copy from the [GitHub repo](#) right now.

Go build it.

Demo Repo

github.com/mistryhardik/community

Global Azure 2026 Mumbai

APIM AI Gateway docs

aka.ms/apim-ai-gateway

Event

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Feedback, questions, and PRs welcome.

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